



Individual Journal

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Group Name: Project Bloom

Role in the Group: Backend Developer, Tester, and Client Developer

Topic / Activity: Laboratory 4: REST API

Date Submitted: April 16, 2026

Overview of My Role

I served as a backend developer, tester, and part-time client-side developer in our project. My main responsibilities included helping design and implement the system's backend functions, testing features to ensure they worked correctly, and assisting with some client-side development when needed. I also worked closely with my groupmates to analyze system issues, identify the causes of bugs, and apply the appropriate fixes during implementation. Through this role, I contributed to improving the functionality, reliability, and overall performance of our system.

Specific Tasks Performed

- Added sample SQL data to support testing and demonstrate how the system works with actual records.
- Contributed partial code in "api.php" to improve functionality and ensure the API worked properly.
- Fixed client-side bugs, especially in the Python-related parts of the system.
- Tested the client side to identify bugs, verify system behavior, and ensure outputs were displayed properly.
- Worked with the team to troubleshoot issues and improve the overall reliability of the system.

Code Contributions

I contributed to the backend by connecting the PHP REST API to the MySQL database using PDO. This part handles the database host, port, database name, username, and password, then creates a reusable connection for the rest of the system. I also added error handling so the API returns a JSON error message if the database connection fails.

I also contributed to the CRUD logic of the system, especially for student management. On the API side, I helped implement the logic for adding student records into the database. On the

JavaScript side, I added client-side actions that call the API to add and remove students, then refresh the interface so the changes appear immediately in the web application.

I also contributed to the Python client by improving the terminal flow so the program is easier to use in the command line. I added terminal clearing and pause handling so each screen is refreshed properly and the output remains organized before returning to the menu.

Challenges Encountered

One of the main challenges I faced was connecting the system to the database. Setting up the database connection and making sure the system could correctly retrieve, store, update, and delete data was difficult, especially during the early stages of development. I had to understand how the backend, API, and database would work together, which required extra effort and careful testing.

Another challenge was my limited knowledge of JavaScript. Since I was not yet very familiar with the language, it was difficult for me to fully understand some parts of the client-side logic and how user interactions were handled in the system. This made debugging and modifying frontend-related features more challenging, and I had to spend additional time learning and reviewing how the code worked.

I also struggled with PHP because I was still not very knowledgeable in it. Since PHP was used in the API and backend development, I found it challenging to understand the syntax, logic flow, and implementation of certain functions. This affected my confidence at first, especially when editing code, fixing errors, or adding features.

Despite these challenges, I was able to improve through practice, testing, and collaboration with my groupmates. By asking questions, reviewing the code carefully, and participating in debugging and implementation, I gradually gained a better understanding of the technologies used in the project. These difficulties became valuable learning experiences that helped me develop my technical skills.

Problem-Solving and Contributions

I helped solve problems in the project by fixing how error messages were displayed and ensuring that the system was connected to the correct database URL. I also searched for online solutions and references to better understand PHP and its implementation in our system. In addition, I explored simple CRUD implementations in different environments, such as Python and JavaScript, to compare approaches and apply ideas that could help improve our project. Through these efforts, I was able to support the group in troubleshooting issues and finding practical solutions during development.